Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

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Abstract

Hyperuniformity describes a state of matter in which large-scale density fluctuations are anomalously suppressed relative to an uncorrelated random arrangement of the same mean density. This suppression is quantified by the hyperuniformity exponent α, which characterizes how the structure factor $S(k)$ vanishes as the wavenumber $k \to 0$, and by the surface-area coefficient $\bar{\Lambda}$, which measures residual local disorder within the strongest (Class I) hyperuniform patterns. Three distinct universality classes exist: Class I ($\alpha > 1$, bounded number variance), Class II ($\alpha = 1$, logarithmic variance), and Class III ($0 < \alpha < 1$, power-law variance).

This work reports on the numerical characterization of 27 one-dimensional point patterns spanning all three classes, carried out under the guidance of Professor Salvatore Torquato at Princeton University. We validate a suite of algorithms for computing number variance, the structure factor, and spreadability against known analytical benchmarks, then apply them to metallic-mean quasicrystals (Fibonacci, Silver, and Bronze chains), perturbed lattices, stealthy hyperuniform configurations, and several aperiodic substitution tilings. We find that all 2×2 metallic-mean substitution matrices yield $\alpha = 3$ by a determinant constraint, leaving the interval $1 < \alpha < 2$ unoccupied by any known quasicrystal. To fill this gap, we analyze the Bombieri-Taylor cubic substitution — a three-letter, 3×3 system whose characteristic polynomial $x^3 - 2x^2 - x + 1 = 0$ breaks the 2×2 constraint — and obtain $\alpha = 1.545$, a Class I quasicrystal with no known predecessor in the literature. We further prove that all 3×3 unimodular Pisot substitution matrices with a complex conjugate eigenvalue pair yield $\alpha = 2$ exactly; combined with an exhaustive numerical search showing no all-real 3×3 case in $(2, 3)$, this confirms that the interval $2 < \alpha < 3$ is inaccessible at algebraic degree three within the row-sum ≤ 4 constraint. An exhaustive search over all 39,304 candidate matrices (of which 4,882 satisfy the Pisot condition) confirms this and yields a new deterministic substitution tiling with $\alpha = 2$ — to our knowledge, the first quasiperiodic substitution chain occupying this value. Generalized scaling metrics are defined for Class II and III patterns. The complete ranking across all 27 patterns establishes a quantitative hierarchy of hyperuniform order and demonstrates that higher-rank matrices are required to engineer exponents in the $2 < \alpha < 3$ range.

Contents

1 Introduction **2**

2 Theoretical Background **3**
2.1 Number Variance 3
2.2 Structure Factor and Hyperuniformity Exponent 4
2.3 Surface-Area Coefficient $\bar{\Lambda}$ 4
2.4 Substitution Tilings and the Eigenvalue Formula 5

3 Methods **5**

1 Introduction

Long-range density fluctuations are a central diagnostic of order in many-body systems. When points are distributed at random with no correlations, the variance of the count inside a window of half-width R grows in proportion to the window length: $\sigma^2_{\text{Poisson}}(R) = 2\rho R$, where ρ is the mean density. This linear growth is the Poisson baseline — the generic behavior of uncorrelated disordered matter. *Hyperuniform* systems are those for which large-scale density fluctuations are anomalously suppressed relative to this baseline [1, 2]. Hyperuniformity is not a statement about short-range structure: a hyperuniform pattern can appear locally disordered while being globally organized in a precise statistical sense. This combination of apparent local randomness with long-range suppression of density fluctuations places hyperuniform systems in a structurally distinct class, intermediate between perfect crystals and uncorrelated fluids.

Hyperuniform arrangements appear across a remarkable range of physical and biological contexts. The photoreceptor mosaic of bird retinas forms a hyperuniform pattern, a design that enables near-uniform visual sampling without the periodic artifacts of a crystal [2]. Systems of hard spheres near their jamming point exhibit hyperuniformity, suggesting that the suppression of density fluctuations is a signature of mechanical criticality. At the largest accessible scales, the distribution of galaxies in cosmological surveys appears nearly hyperuniform, hinting at long-range correlations established in the early universe. Quasicrystals — aperiodic materials with long-range orientational order but no translational periodicity, first observed in an aluminium-manganese alloy by Shechtman and coworkers [3] — are naturally hyperuniform because their long-range aperiodic structure prevents macroscopic density imbalances. From an engineering perspective, computationally designed disordered hyperuniform networks have been shown to support photonic band gaps rivaling those of periodic photonic crystals, offering a route to isotropic wave-guiding that does not require strict periodicity [2].

The formal definition of hyperuniformity, due to Torquato and Stillinger [1], is that the structure factor $S(k)$ — the Fourier-space measure of density fluctuations at spatial wavelength $2\pi/k$ — satisfies $S(k) \to 0$ as $k \to 0$. The *rate* of this vanishing defines a spectrum of hyperuniformity. For systems where $S(k) \sim |k|^\alpha$ near $k = 0$, the exponent $\alpha > 0$ quantifies the degree of suppression: larger α means faster vanishing and more strongly reduced fluctuations at long wavelengths. Three universality classes arise from the asymptotic behavior of $\sigma^2(R)$ [2]. Class I ($\alpha > 1$) is the strongest: the number variance is bounded at large R, approaching a constant plateau. Perfect crystals and many quasicrystals belong to this class, as do stealthy hyperuniform systems for which $S(k) = 0$ over an entire exclusion zone near $k = 0$. Class II ($\alpha = 1$) is the marginal case where $\sigma^2(R)$ grows logarithmically; the period-doubling substitution chain is a one-dimensional example. Class III ($0 < \alpha < 1$) gives power-law variance growth strictly slower than the Poisson baseline; certain substitution chains with non-Pisot eigenvalues fall here.

One spatial dimension is a natural test bed for hyperuniformity theory. Point patterns in 1D can be generated deterministically to sizes $N \sim 10^7$ by iterating substitution rules, exact analytical benchmarks exist, and the connection between a pattern's algebraic structure and its hyperuniformity class is particularly transparent. Yet the 1D landscape of known Class I patterns is sparse. The survey in Ref. [2] lists only a handful of examples, and the interval $1 < \alpha < 2$ was previously unoccupied by any known 1D quasicrystal. As we show below, the reason is algebraic: all 2×2 metallic-mean substitution matrices are unimodular ($|\det M| = \pm 1$), which forces $\alpha = 3$ exactly [4]. Filling the $1 < \alpha < 2$ gap requires moving outside this family.

In this work, we characterize 27 one-dimensional point patterns spanning all three hyperuniformity classes. The patterns include the five metallic-mean quasicrystals (Fibonacci, Silver, Bronze, Copper, and Nickel chains), all of which have $\alpha = 3$ by the determinant constraint; uncorrelated

random lattice (URL) perturbations, which realize $\alpha = 2$ exactly and serve as key validation benchmarks; stealthy hyperuniform configurations generated by collective coordinate optimization; and several substitution chains that realize Class II and Class III behavior. The central new result is the identification of the Bombieri-Taylor cubic substitution [5] as a Class I quasicrystal with $\alpha = 1.545$, the first known example with $1 < \alpha < 2$ [4]. We also report novel values of $\bar{\Lambda}$ for the Silver, Bronze, Copper ($\bar{\Lambda} = 0.293$), and Nickel ($\bar{\Lambda} = 0.310$) chains; the last two are reported here for the first time.

The remainder of this paper is organized as follows. Section 2 develops the theoretical framework: the number variance and its relation to the pair correlation function, the structure factor and hyperuniformity exponent, the surface-area coefficient $\bar{\Lambda}$, and the eigenvalue formula for substitution tilings. Section 3 describes the numerical algorithms and validation benchmarks. Results sections will follow.

2 Theoretical Background

2.1 Number Variance

To quantify how density fluctuations depend on scale, we examine how many points fall inside a window of half-width R placed at a random position along a one-dimensional point pattern of mean linear density ρ. The variance of this count across window placements is the *number variance* $\sigma^2(R)$. For a Poisson (completely uncorrelated) point process with density ρ, the variance grows linearly with window size: $\sigma^2_{\text{Poisson}}(R) = 2\rho R$ [1]. This Poisson scaling is the baseline against which all other patterns are measured — it reflects the fact that in an uncorrelated system, each region of space is independent, so larger windows simply accumulate more independent uncertainty.

For a general stationary ergodic point process, $\sigma^2(R)$ can be expressed in terms of the pair correlation function $g_2(r)$, which encodes the probability of finding a second point at distance r from a reference point relative to the uncorrelated expectation [1]:

$$\sigma^2(R) = 2\rho R \left[1 + 2\rho \int_0^{2R} \left(1 - \frac{r}{2R}\right) \left[g_2(r) - 1\right] dr \right]. \tag{1}$$

The first term, $2\rho R$, is the uncorrelated (Poisson) contribution. The integral represents the correction from inter-point correlations: where $g_2(r) > 1$ (clustering), the variance increases; where $g_2(r) < 1$ (anticorrelation), the variance decreases. A pattern is **hyperuniform** if the integral correction exactly cancels the Poisson term in the large-R limit [1]:

$$\lim_{R\to\infty} \frac{\sigma^2(R)}{2\rho R} = 0. \tag{2}$$

Physically, this requires that points be anticorrelated at large separations in precisely the right amount: $\int_0^\infty [g_2(r) - 1]\, dr = -1/(2\rho)$.

Based on how $\sigma^2(R)$ grows asymptotically, hyperuniform patterns fall into three universality classes [2]:

- **Class I** ($\alpha > 1$): $\sigma^2(R)$ approaches a finite (though possibly oscillating) constant at large R. These patterns exhibit the strongest suppression of density fluctuations. Perfect crystals and quasicrystals generated by Pisot substitution rules belong here.

- **Class II** ($\alpha = 1$): $\sigma^2(R) \sim C \ln R$. The variance grows without bound, but only logarithmically. The period-doubling chain is a 1D example.

- **Class III** ($0 < \alpha < 1$): $\sigma^2(R) \sim R^{1-\alpha}$, a power-law growth strictly slower than the Poisson baseline. Aperiodic chains with non-Pisot eigenvalue spectra can realize this behavior.

Non-hyperuniform patterns ($\alpha \leq 0$) have $\sigma^2(R) \sim R$ or faster.

2.2 Structure Factor and Hyperuniformity Exponent

The same information encoded in $\sigma^2(R)$ is also captured by the *structure factor* in Fourier space. For a point pattern with N points at positions $\{x_j\}$,

$$S(k) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{ikx_j} \right|^2 . \tag{3}$$

Physically, $S(k)$ measures the intensity of density fluctuations at spatial wavelength $2\pi/k$. For a Poisson process $S(k) = 1$ at all k; for a perfect crystal $S(k) = 0$ for all $|k|$ below the first reciprocal lattice vector. The two observables are related by an exact integral [1]:

$$\sigma^2(R) = \frac{\rho}{\pi} \int_0^\infty S(k) \left(\frac{2\sin(kR)}{k} \right)^2 dk. \tag{4}$$

Hyperuniformity requires $S(k) \to 0$ as $k \to 0$, which causes the integrand to vanish faster than for Poisson and yields sub-linear variance growth at large R.

For systems where $S(k)$ approaches zero as a power law,

$$S(k) \sim B|k|^\alpha \quad \text{as } k \to 0, \tag{5}$$

the exponent $\alpha > 0$ is the *hyperuniformity exponent*. Larger α implies more strongly suppressed fluctuations at long wavelengths. For quasicrystals, the structure factor consists of dense Bragg peaks rather than a smooth continuous envelope, so the exponent is extracted from the integrated intensity $Z(k) = 2\int_0^k S(q)\,dq \sim k^{1+\alpha}$ near $k = 0$ [4]. In practice, α is also estimated from the asymptotic slope of $\sigma^2(R)$ on a log-log scale, providing an independent numerical check.

2.3 Surface-Area Coefficient $\bar{\Lambda}$

Within Class I, all patterns have bounded number variance, but two systems with the same α can differ substantially in their residual local disorder. The *surface-area coefficient* $\bar{\Lambda}$ is the secondary metric that resolves this ambiguity and provides an absolute ranking within Class I.

For Class I systems in one dimension, $\sigma^2(R)$ does not simply converge to a constant but typically oscillates about a plateau, particularly for quasicrystals with dense Bragg spectra. A pointwise limit is therefore not well-defined in general. Instead, $\bar{\Lambda}$ is defined as the long-range running average [2, 1]:

$$\bar{\Lambda} = \lim_{R\to\infty} \frac{1}{R} \int_0^R \sigma^2(R')\,dR'. \tag{6}$$

This running average converges to the mean level of the oscillating variance plateau, and reduces to the ordinary limit $\lim_{R\to\infty} \sigma^2(R)$ whenever that limit exists. For systems where the variance oscillates without converging, Eq. (6) extracts a unique, well-defined amplitude.

The name "surface-area coefficient" refers to the d-dimensional interpretation: for a Class I pattern in d dimensions, the variance scales at large R as $\sigma^2(R) \sim 2\bar{\Lambda}\,R^{d-1}$, where R^{d-1} is proportional to the surface area of the observation window. In $d = 1$ this reduces to $\sigma^2(R) \to \bar{\Lambda}$ (a

constant), consistent with the definition Eq. (6). The quantity $\bar{\Lambda}$ is the primary ranking metric within Class I: the perfect integer lattice achieves the global minimum $\bar{\Lambda} = 1/6$, proved exact in Ref. [1], and any deviation from perfect periodicity strictly increases $\bar{\Lambda}$. For the Fibonacci chain, Ref. [6] established $\bar{\Lambda} \approx 0.201$.

2.4 Substitution Tilings and the Eigenvalue Formula

A one-dimensional *substitution tiling* is an aperiodic sequence of tiles generated by repeatedly applying a fixed rewriting rule to each letter of a finite alphabet. Starting from a seed word, each letter is replaced by a fixed string of letters; iteration produces an increasingly long sequence that, when each tile is assigned a physical length and the tiles are stacked end to end, defines a one-dimensional point pattern. These tilings provide some of the cleanest examples of quasicrystalline order: they are fully deterministic, can be generated to arbitrary length by iteration, and their algebraic properties encode their hyperuniformity class in a precise and computable way.

The growth of a substitution tiling is governed by its *substitution matrix* M, whose entry M_{ij} counts how many tiles of type j appear in the substituted image of a single tile of type i. For the Fibonacci chain with alphabet $\{S, L\}$ and rules $S \to L$, $L \to LS$, the substitution matrix is

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \tag{7}$$

with eigenvalues $\lambda_1 = \tau = (1 + \sqrt{5})/2 \approx 1.618$ (the golden ratio) and $\lambda_2 = -1/\tau \approx -0.618$. The Perron-Frobenius theorem guarantees that the largest eigenvalue λ_1 is positive and real, and its eigenvector gives the asymptotic relative frequencies of each tile type. Tile lengths are assigned by requiring that inflating every tile by λ_1 and then applying the substitution leaves the total domain length invariant, which fixes the ratio $L/S = \lambda_1$. For the Fibonacci chain this gives $L = \tau S$: the long tile is τ times the short tile.

The key result connecting the matrix spectrum to hyperuniformity is due to Oğuz et al. [7, 4]: for a substitution tiling with Perron-Frobenius eigenvalue λ_1 and second-largest-magnitude eigenvalue λ_2,

$$\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln |\lambda_1|}. \tag{8}$$

This formula is the central predictive tool of this work: α is determined entirely by the substitution matrix spectrum, with no simulation required. The tiling is hyperuniform when $\alpha > 0$, which requires $|\lambda_2| < |\lambda_1|^{1/2}$. The physical interpretation is clear: $|\lambda_1|$ sets the inflation rate under substitution, while $|\lambda_2|$ controls how quickly spatial correlations across tile boundaries decay. A larger ratio $|\lambda_2|/|\lambda_1|^{1/2}$ means slower decorrelation and weaker hyperuniformity.

For the metallic-mean chains and for the Bombieri-Taylor cubic substitution, the largest eigenvalue λ_1 is a *Pisot-Vijayaraghavan (PV) number*: an algebraic integer greater than one all of whose algebraic conjugates lie strictly inside the unit circle ($|\lambda_j| < 1$ for $j \geq 2$) [5, 4]. The PV property guarantees a pure Bragg diffraction spectrum — the tiling is a true quasicrystal in the crystallographic sense, with delta-function peaks in $S(k)$ at a dense set of wavevectors. Equation (8) also reveals the structural reason why all 2×2 metallic-mean matrices are locked to $\alpha = 3$ and why accessing intermediate exponents requires a higher-rank substitution matrix.

3 Methods

The hyperuniformity of each pattern is characterized numerically using three complementary observables: the number variance $\sigma^2(R)$, the structure factor $S(k)$, and the spreadability $\mathcal{S}(t)$ (the

fraction of space not yet covered by a diffusing species at time t, whose long-time decay rate probes the same fluctuation spectrum as $S(k)$ [8, 9]). Each observable is computed from a finite point pattern generated to a sufficiently large number of points N that finite-size effects are negligible on the window scales of interest.

Number variance. For a pattern containing N points on a domain of length L, the variance is estimated by averaging the squared fluctuation in point counts over $N_{\text{samp}} = 30{,}000$ window placements chosen uniformly at random along the domain. The window half-width R is varied from $R_{\min} = 1/\rho$ to $R_{\max} = L/4$ to avoid boundary effects. For substitution-tiling patterns, $N \sim 10^4$–10^7 depending on the convergence requirements; for stealthy patterns, $N = 2000$ points were provided per configuration and approximately 4000 independent configurations were averaged to reduce stochastic noise.

Structure factor. The structure factor is computed as

$$S(k) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{ikx_j} \right|^2 , \tag{9}$$

evaluated on the discrete set of wavenumbers $k = 2\pi n/L$ for integers n. The low-k regime is sampled with fine resolution and binned logarithmically to reduce noise. The exponent α is extracted from an unweighted linear fit to $\log S(k)$ versus $\log k$ over a range of k values where power-law behavior is evident.

Spreadability. The spreadability $\mathcal{S}(t)$ is evaluated via the integral representation involving $S(k)$, using a fast summation over the precomputed structure factor. Its long-time behavior provides an independent consistency check on α.

Validation. Before applying these methods to the patterns under study, all three algorithms are validated against exact benchmarks: the Poisson process ($\sigma^2 = 2\rho R$, $S(k) = 1$) and the uncorrelated random lattice, for which $\bar{\Lambda} = (1 + a^2)/6$ is known analytically [10]. Poisson benchmark agreement was confirmed to within 1.1% error; URL agreement to within 0.3%.

Substitution-tiling generation. Quasicrystal patterns are generated by iterating the substitution rule until the total point count exceeds a target N_{target}. Tile lengths are assigned from the Perron-Frobenius eigenvector of M, normalized so that the shortest tile has length one. Points are placed at tile boundaries, yielding a one-dimensional point process with mean density $\rho = 1/\bar{\ell}$, where $\bar{\ell}$ is the mean tile length weighted by asymptotic tile frequencies.

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